

TidyTuesday Week 32

Pranay Gundam

Wednesday 5th August, 2026

Contents

1	Weekly Summary	2
2	Date: 2026-08-03	3
2.1	Raw Regression	3
2.2	Detrended Regression	5
3	Date: 2026-08-04	7
3.1	Raw Regression	7
3.2	Detrended Regression	9
4	Date: 2026-08-05	11
4.1	Raw Regression	11
4.2	Detrended Regression	13

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the segment where we heroically pair up economic time series that have never met, force them into a regression, and act shocked at the results. This week the random number generator was feeling especially whimsical, so buckle up.

Monday gave us the crown jewel: Coinbase Bitcoin versus the San Francisco Tech Pulse, a tech index so vibrant it was discontinued in 2020. The raw regression came in with an R-squared of about 0.70, which in this project counts as basically a spiritual experience – seventy percent! A p-value with seventeen zeros! For a moment I genuinely thought we'd cracked the market. And then, as always, we ran the detrended version, where the shared "everything went up over time" story gets stripped out, and the R-squared face-planted from 0.70 all the way down to 0.14. So yes, Bitcoin and a defunct tech index both trended upward for years. Groundbreaking. This is the exact trend-mirage the project loves to point at and laugh: two lines going up together is not romance, it's just gravity in reverse. To its modest credit, the detrended version was still technically "significant," but at that R-squared I wouldn't invite it to dinner.

Tuesday was the palate cleanser of pure nothing: Philadelphia's Current Physical Plant Capital Expenditures diffusion index against the Real User Cost Index of MSI-M2M, a name I am contractually obligated to pretend I understand. The raw regression posted an R-squared of 0.0006 and a p-value of 0.89, which is statistics-speak for "these two have never been in the same room." The detrended version somehow did even less, if that's possible. No trend mirage, no relationship, no drama – just two series shrugging at each other across the void. Beautiful, honestly. This is what most days actually look like.

Wednesday brought Banca d'Italia's foreign exchange interventions from the '90s versus the delinquency rate on real estate loans at the 100 largest banks, because of course it did. The raw regression squeaked in "significant" at a p-value of 0.04 with a whopping 9.5 percent R-squared, the kind of result that gets you a participation trophy. Then we detrended it and the p-value ballooned to 0.35, meaning the relationship evaporated the instant we took away its trendy little crutch. Another mirage identified and dismissed. As always, please remember that everything here is randomly paired series with zero causal claim and even less statistical rigor – spurious correlation isn't the bug, it's the entire personality of this project. See you next week, when the dice roll again.

2 Date: 2026-08-03

Series ID: CBBTCUSD

This series is titled Coinbase Bitcoin and has a frequency of Daily, 7-Day. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2014-12-01 and the observation end date is 2026-08-02. The popularity of this series is 75.

Series ID: SFTPINDM114SFRBSF

This series is titled San Francisco Tech Pulse (DISCONTINUED) and has a frequency of Monthly. The units are Index Jan 2000=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1971-04-01 and the observation end date is 2020-03-01. The popularity of this series is 10.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF	R-squared:	0.695
Model:	OLS	Adj. R-squared:	0.690
Method:	Least Squares	F-statistic:	139.1
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	2.21e-17
Time:	14:09:51	Log-Likelihood:	-185.92
No. Observations:	63	AIC:	375.8
Df Residuals:	61	BIC:	380.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	84.9060	0.863	98.348	0.000	83.180	86.632
value_fred_CBBTCUSD	0.0018	0.000	11.796	0.000	0.002	0.002

Omnibus:	9.975	Durbin-Watson:	0.305
Prob(Omnibus):	0.007	Jarque-Bera (JB):	10.825
Skew:	0.715	Prob(JB):	0.00446
Kurtosis:	4.442	Cond. No.	8.11e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.11e+03. This might indicate that there are strong multicollinearity or other numerical problems.

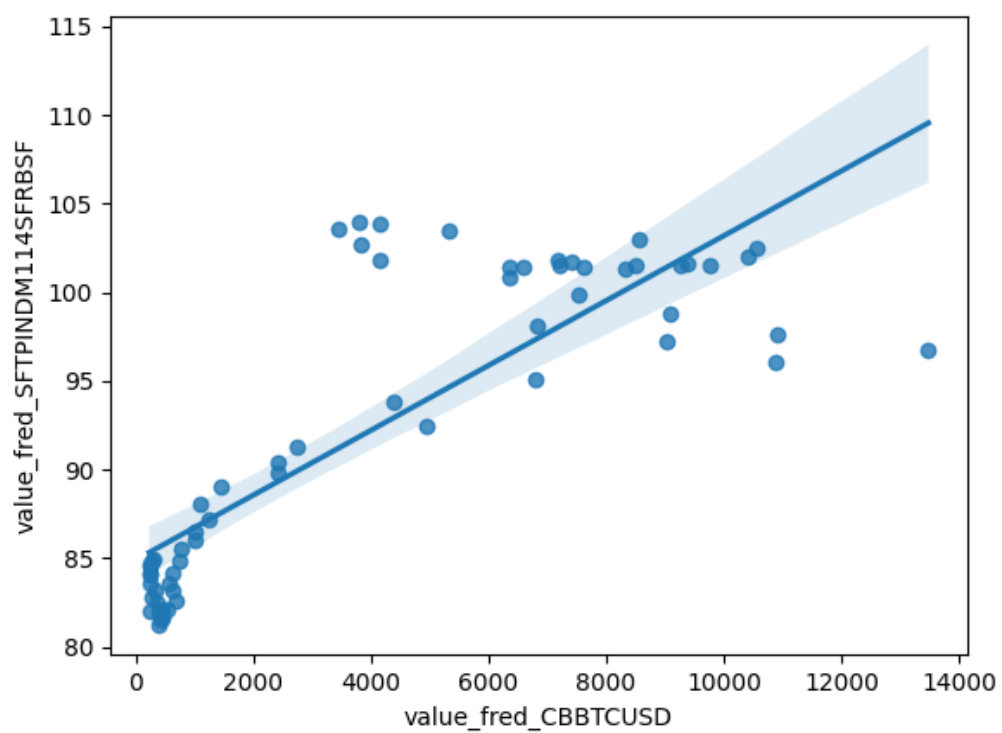


Figure 1: Raw Regression Plot for 2026-08-03

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF_detrended	R-squared:	0.137
Model:	OLS	Adj. R-squared:	0.123
Method:	Least Squares	F-statistic:	9.672
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	0.00284
Time:	14:09:51	Log-Likelihood:	-150.74
No. Observations:	63	AIC:	305.5
Df Residuals:	61	BIC:	309.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.857e-12	0.339	1.14e-11	1.000	-0.678	0.678
value_fred_CBBTCUSD_detrended	0.0005	0.000	3.110	0.003	0.000	0.001

Omnibus:	23.048	Durbin-Watson:	0.095
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4.918
Skew:	0.258	Prob(JB):	0.0855
Kurtosis:	1.732	Cond. No.	2.25e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

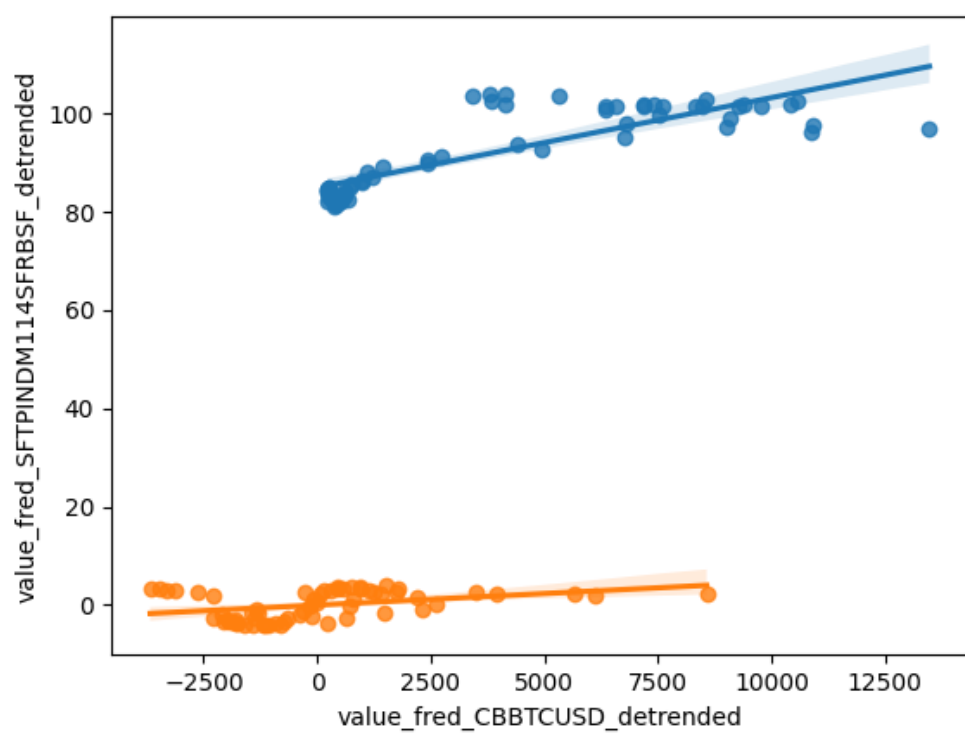


Figure 2: Detrended Regression Plot for 2026-08-03

3 Date: 2026-08-04

Series ID: CPBNDIF066MSFRBPHI

This series is titled Current Physical Plant Capital Expenditures; Diffusion Index for Federal Reserve District 3: Philadelphia and has a frequency of Monthly. The units are Index and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2011-03-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

Series ID: OCM2MP

This series is titled Real User Cost Index of MSI-M2M (preferred) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1967-01-01 and the observation end date is 2013-12-01. The popularity of this series is 4.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OCM2MP	R-squared:	0.001
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.01870
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.892
Time:	13:31:41	Log-Likelihood:	184.86
No. Observations:	34	AIC:	-365.7
Df Residuals:	32	BIC:	-362.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.0155	0.000	36.337	0.000	0.015	0.016
value_fred_CPBNDIF066MSFRBPHI	3.385e-06	2.47e-05	0.137	0.892	-4.7e-05	5.38e-05

Omnibus:	9.983	Durbin-Watson:	0.076
Prob(Omnibus):	0.007	Jarque-Bera (JB):	2.548
Skew:	0.142	Prob(JB):	0.280
Kurtosis:	1.689	Cond. No.	39.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

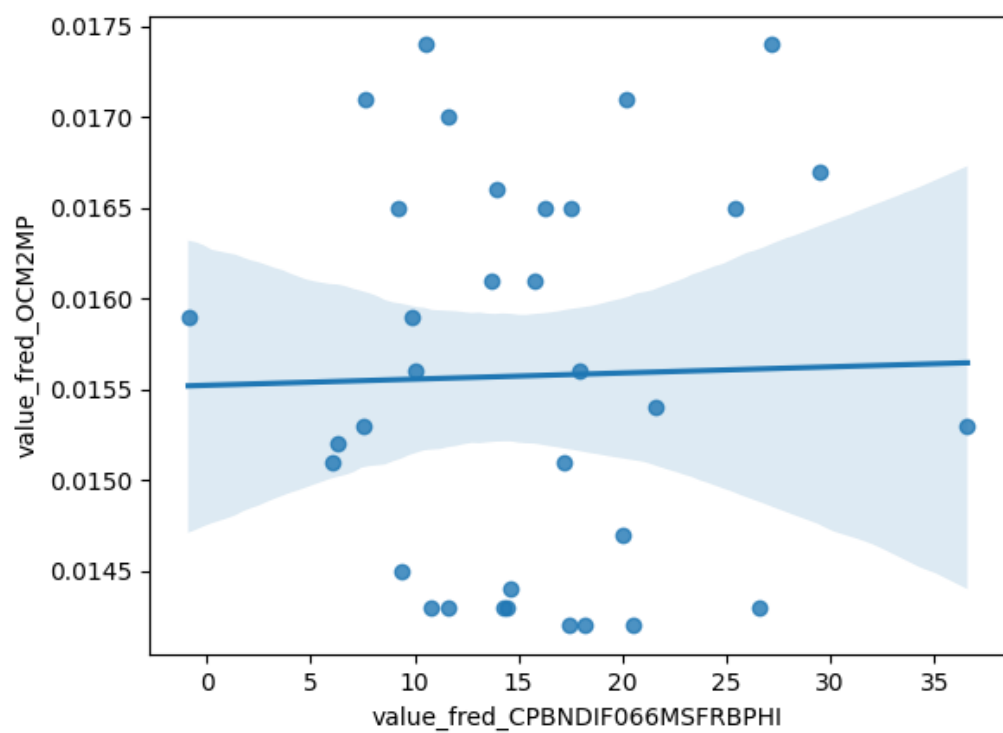


Figure 3: Raw Regression Plot for 2026-08-04

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OCM2MP_detrended	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.006047
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.939
Time:	13:31:41	Log-Likelihood:	193.17
No. Observations:	34	AIC:	-382.3
Df Residuals:	32	BIC:	-379.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.858e-15	0.000	3.33e-11	1.000	-0.000	0.000
value_fred_CPBNDIF066MSFRBPHI_detrended	1.508e-06	1.94e-05	0.078	0.939	-3.8e-05	4.1e-05

Omnibus:	10.150	Durbin-Watson:	0.122
Prob(Omnibus):	0.006	Jarque-Bera (JB):	2.467
Skew:	0.016	Prob(JB):	0.291
Kurtosis:	1.681	Cond. No.	7.52

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

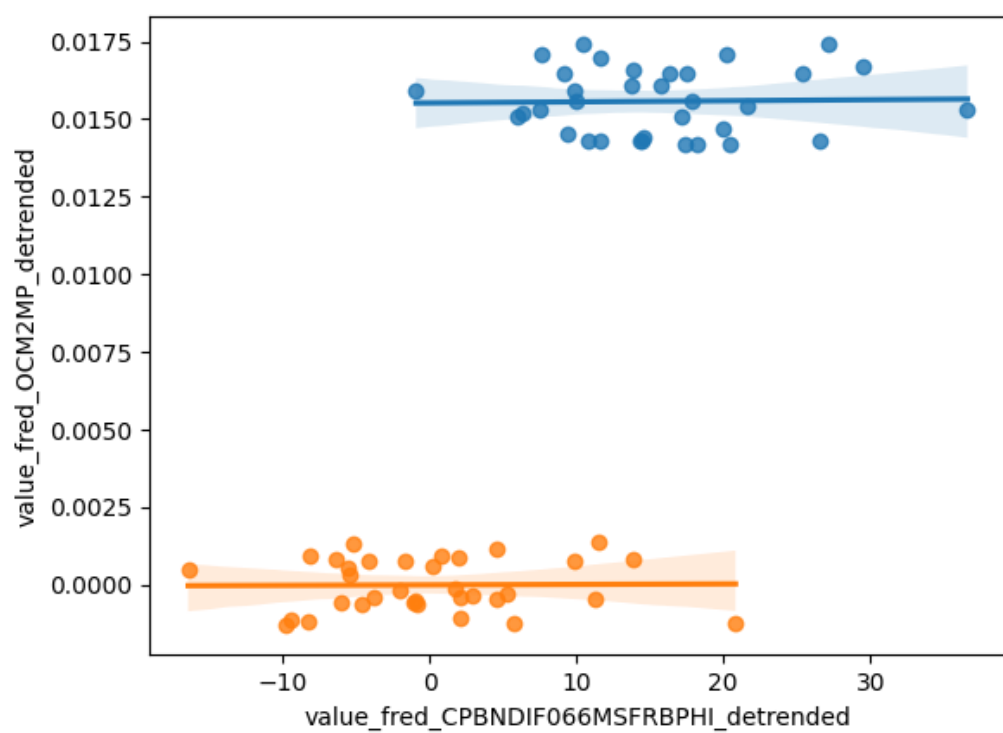


Figure 4: Detrended Regression Plot for 2026-08-04

4 Date: 2026-08-05

Series ID: ITINTDRES

This series is titled Italian Intervention: Banca d'Italia Purchases of Foreign Exchange (Millions of ECU's, Euros after 1999) (DISCONTINUED) and has a frequency of Daily, 7-Day. The units are Millions of ECU's, Euros after 1999 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 1998-12-31. The popularity of this series is 2.

Series ID: DRSRET100N

This series is titled Delinquency Rate on Loans Secured by Real Estate, Banks Ranked 1st to 100th Largest in Size by Assets and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1987-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

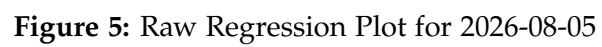
4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DRSRET100N	R-squared:	0.095			
Model:	OLS	Adj. R-squared:	0.074			
Method:	Least Squares	F-statistic:	4.433			
Date:	Wed, 05 Aug 2026	Prob (F-statistic):	0.0413			
Time:	13:32:37	Log-Likelihood:	-100.49			
No. Observations:	44	AIC:	205.0			
Df Residuals:	42	BIC:	208.5			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	5.6231	0.377	14.905	0.000	4.862	6.384
value_fred_ITINTDRES	-0.0081	0.004	-2.105	0.041	-0.016	-0.000
	Omnibus:	5.847	Durbin-Watson:	0.227		
	Prob(Omnibus):	0.054	Jarque-Bera (JB):	2.476		
	Skew:	0.255	Prob(JB):	0.290		
	Kurtosis:	1.956	Cond. No.	101.		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DRSRET100N_detrended	R-squared:	0.021
Model:	OLS	Adj. R-squared:	-0.002
Method:	Least Squares	F-statistic:	0.9112
Date:	Wed, 05 Aug 2026	Prob (F-statistic):	0.345
Time:	13:32:38	Log-Likelihood:	-87.537
No. Observations:	44	AIC:	179.1
Df Residuals:	42	BIC:	182.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-5.314e-13	0.273	-1.95e-12	1.000	-0.551	0.551
value_fred_ITINTDRES_detrended	-0.0029	0.003	-0.955	0.345	-0.009	0.003

Omnibus:	3.670	Durbin-Watson:	0.127
Prob(Omnibus):	0.160	Jarque-Bera (JB):	3.038
Skew:	0.532	Prob(JB):	0.219
Kurtosis:	2.275	Cond. No.	90.8

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

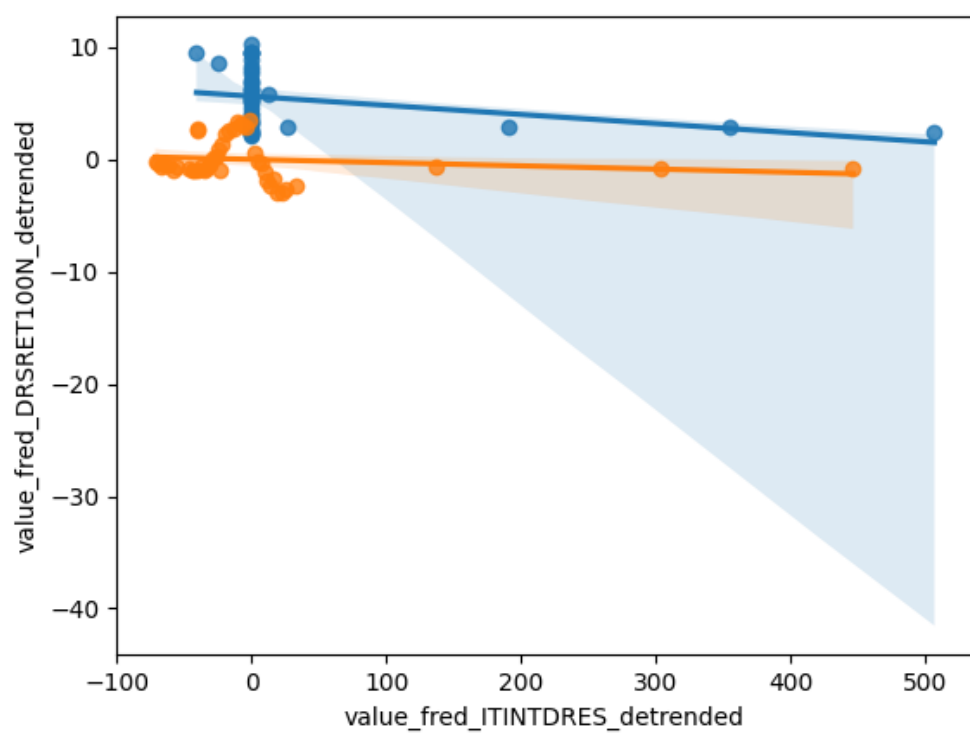


Figure 6: Detrended Regression Plot for 2026-08-05